

PAPER_448 — Multi-System UQFF Core Compression Framework: Unified F_env Architecture

Whitepaper Series: Star-Magic UQFF Phase 2

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Classification: FIRST unified multi-system F_env modular architecture in UQFF; FIRST std::map dynamic variable storage for astrophysical UQFF systems

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Abstract

The Multi-System UQFF Core Compression Framework establishes the architectural template for UQFF Compression Cycle 2, defining the shared methodology by which an arbitrary number of astrophysical systems can be simultaneously calculated under a single compressed gravitational equation. The unified architecture uses modular environmental factor (F_env) summation with Hubble evolution $H(t, z)$, standard map-based dynamic variable storage, and consistent UQFF/MUGE term parameterisation across all Cycle 2 systems. The seven canonical systems (MagnetarSGR1745, SagittariusA, TapestryStarbirth, Westerlund2, PillarsCreation, RingsRelativity, UniverseGuide) define the baseline registry from which the 19- and 29-system expansions are derived.

2. Core Architecture — PAPER_448

2.1 Compression Cycle 2 Philosophy

In UQFF Compression Cycle 1 (Sessions 1-114), each astrophysical system maintained a dedicated class with hardcoded parameters. Compression Cycle 2 introduces a **generalised modular registry** in which:

1. System parameters are stored in `std::map<std::string, double>` containers
2. Environmental factors F_env are computed as a unified sum across all system-specific terms
3. A single compressed $g_{UQFF}(t)$ equation applies to any registered system
4. Dynamic parameter updates propagate automatically through the registry

2.2 Unified Gravitational Equation (Core)

$$g_{UQFF}(r, t) = \frac{GM(t)}{r^2} (1 + H_z t) (1 - B/B_{crit}) + \sum_i U_{gi} + \frac{\Lambda c^2}{3} + g_{quantum} + g_{fluid} + F_{env}(t)$$

Where the total environmental modifier:

$$F_{\text{env}}(t) = \sum_{j=1}^{N_{\text{sys}}} F_{\text{env}}^{(j)}(t, \{p_j\})$$

With $\{p_j\}$ = system-specific parameter map for system j .

2.3 Hubble Evolution Module

$$H(t, z) = H_0 \sqrt{\Omega_m (1+z)^3 + \Omega_\Lambda}$$

Evaluated at z for each system: - Local ($z \approx 0$): $H \approx 70$ km/s/Mpc - Intermediate ($z=0.5$): $H \approx 85$ km/s/Mpc
- Cosmological ($z=1100$, CMB): $H \approx 70 \times \sqrt{(0.3 \times 1100^3 + 0.7)}$ km/s/Mpc

$$H_z = H(z)/H_0 \text{ [dimensionless Hubble factor]}$$

2.4 Dynamic Variable Storage (FIRST in UQFF)

The C++ implementation introduces `std::map<std::string, double>` as the canonical storage for per-system variables:

```
params["M"]      = system_mass      [kg]
params["r"]      = radius           [m]
params["z"]      = redshift         [dimensionless]
params["B"]      = magnetic_field   [T]
params["v_exp"]  = expansion_vel    [m/s]
params["rho"]    = fluid_density    [kg/m³]
params["F_env"]  = env_modifier     [m/s²]
params["SC_m"]   = superconductive  [dimensionless]
```

This is the **first use of runtime-keyed variable maps** for per-system UQFF gravity in the codebase — replacing class-member variables with $O(\log n)$ lookup tables allowing unlimited parameter extension without recompilation.

2.5 Canonical 7-System Registry

System	type_key	key parameters
MagnetarSGR1745	MAGNETAR_SGR1745	$M=2.8 M_\odot$, $r=1e4$ m, $B=1e11$ T
SagittariusA	SAGITTARIUS_A	$M=4.1e6 M_\odot$, $r=6e9$ m, $B=1e-3$ T
TapestryStarbirth	TAPESTRY_STARBIRTH	$M=500 M_\odot$, $r=1e16$ m, $z=0.001$
Westerlund2	WESTERLUND2	$M=1e4 M_\odot$, $r=6e16$ m, $z=0.005$
PillarsCreation	PILLARS_CREATION	$M=200 M_\odot$, $r=6e16$ m, $z=0.002$
RingsRelativity	RINGS_RELATIVITY	$M=1e39$ kg, $r=1e20$ m, $z=0.3$
UniverseGuide	UNIVERSE_GUIDE	$M=1e53$ kg, $r=4.4e26$ m, $z=1100$

These 7 are the Compression Cycle 2 **root systems** — all subsequent 19- and 29-system registries add to this base.

3. Ug Component Summary

3.1 Ug1 — Magnetic Dipole Term

$$U_{g1} = -\frac{\mu_0 m_1 m_2}{4\pi r^3} \left(1 - \frac{r_s}{r}\right)$$

3.2 Ug2 — Charge Reactivity Term

$$U_{g2} = k_e \frac{q_1 q_2}{r} \left(1 + \frac{\kappa t}{\tau_q}\right)$$

3.3 Ug3 — String Rotation Term (UQFF-specific)

$$U_{g3} = \frac{GM_{\text{ext}}}{r_{\text{ext}}^2} \left(1 + \frac{v_s}{c} \cos \theta\right)$$

Compressed form (Cycle 2):

$$U'_{g3} = \frac{GM_{\text{ext}}}{r_{\text{ext}}^2}$$

3.4 Ug4 — Vacuum Concentration Term

$$U_{g4} = U_A \rho_{\text{vac}} (1 + [\text{UA}] \cdot [\text{SCm}])$$

4. Quantum and Fluid UQFF Terms

4.1 Quantum Gravity Coupling

$$g_{\text{quantum}} = \frac{\hbar c}{l_p^2} \frac{t}{t_p} \cdot \frac{1}{Mc^2}$$

Where $l_p = \sqrt{\hbar G/c^3} = 1.616 \times 10^{-35}$ m (Planck length), $t_p = l_p/c$.

4.2 Fluid Navier-Stokes Coupling

$$g_{\text{fluid}} = \nu_{\text{fluid}} \nabla^2 v + (\mathbf{v} \cdot \nabla) \mathbf{v}$$

Simplified for radial symmetry:

$$g_{\text{fluid}} \approx \rho_{\text{fluid}} v_{\text{exp}}^2 / r$$

5. Ψ_{total} Integration

The full quantum-gravitational wave function total combines UQFF modes:

$$\psi_{\text{total}} = \int_0^\infty A(k) e^{i(kr - \omega t)} dk + \frac{[SSq]^{n_{26}}}{[SSq]^{n_{26}-1}}$$

The discrete multi-system version sums over all N registered systems:

$$\psi_{\text{total}}^{(N)} = \sum_{j=1}^N g_{\text{UQFF}}^{(j)}(r, t) \cdot w_j$$

Where w_j = system weight (default: equal weights = 1/N).

6. Standard Model Comparison

Feature	SM	CC2 (UQFF)
Per-system gravity	Individual Poisson eq.	Unified compressed registry
Environmental coupling	External perturbation	Built-in <code>F_env</code> term
Dynamic parameter storage	Compile-time	Runtime <code>std::map</code>
Multi-system simultaneity	Separate solvers	Single <code>g_UQFF</code> call for N systems

7. Testable Predictions

1. **Runtime extensibility:** Adding a new astrophysical system to the Cycle 2 registry should require zero recompilation — only map insertion. Testable by adding any new entry and verifying output.
2. **`F_env` additivity:** For two weakly-interacting systems (e.g., Tapestry + Pillars at large separation), $F_{\text{env_total}} \approx F_{\text{env_1}} + F_{\text{env_2}}$ within 0.1%.

3. **Hubble evolution consistency:** $H(z=0.5)$ from the modular $H(t,z)$ function should reproduce $H_0\sqrt{0.3 \times 1.5^3 + 0.7} = H_0 \times 0.894 = 62.6 \text{ km/s/Mpc } (\pm 1\%)$.

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: $\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$	$\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$ (PDG QED exact)	PDG 2024	100% (exact QED input)
Astrophysical system luminosity X-ray / Radio	UQFF MUGE $g_{\text{total}} \rightarrow L_X$ via Stefan-Boltzmann + buoyancy flux: $L_X \approx g_{\text{total}} \times M_{\text{env}}$	$L_X L \geq 10^{37} \text{ erg/s}$	Chandra CXC	□ Consistent order of magnitude
GR Schwarzschild limit	UQFF g_{total} must satisfy $g \leq c^2/(2r_s)$ at event horizon	$r_s = 2GM/c^2$ (GR exact)	PDG 2024 / GR	□ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$	Observed X-ray variability τ_{obs} (instrument monitoring)	Chandra CXC	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Astrophysical system through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

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